

NOTES: CLASS-9

Complex form of Fourier series

The complex exponential form of the Fourier series is

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{i n \pi x}{l}} \text{ where } c_n = \frac{1}{2l} \int_c^{c+2L} f(x) e^{\frac{-i n \pi x}{l}} dx.$$

Problem 1. Find the complex form of the Fourier Series of $f(x) = e^{-x} \quad -1 \leq x \leq 1$

Solution : Here period, $2l = 2 \Rightarrow l = 1$

The complex form of The Fourier Series of $f(x)$ is given by,

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i n \pi x / l}$$

where

$$\begin{aligned} c_n &= \frac{1}{2l} \int_{-l}^l f(x) e^{-i n \pi x / l} dx. \\ &= \frac{1}{2} \int_{-1}^1 e^{-x} e^{-i n \pi x / 1} dx \\ &= \frac{1}{2} \int_{-1}^1 e^{-(i n \pi + 1)x} dx \\ &= \frac{1}{2} \left[\frac{e^{-(i n \pi + 1)x}}{-(i n \pi + 1)} \right]_{-1}^1 \\ &= \frac{e^{-(i n \pi + 1)} - e^{(i n \pi + 1)}}{-2(i n \pi + 1)} \\ &= \frac{e^{(i n \pi + 1)} - e^{-(i n \pi + 1)}}{2(i n \pi + 1)} \\ &= \frac{e^1 \cdot e^{i n \pi} - e^{-1} \cdot e^{-i n \pi + 1}}{2(i n \pi + 1)} \\ &= \frac{e(\cos n \pi + i \sin n \pi) - e^{-1}(\cos n \pi + i \sin n \pi)}{2(i n \pi + 1)} \\ &= \frac{e \cdot \cos n \pi - e^{-1} \cdot \cos n \pi}{2(i n \pi + 1)} \\ &= \frac{(e - e^{-1})(-1)^n}{2(i n \pi + 1)} \\ &= \frac{(-1)^n \sinh 1}{(i n \pi + 1)} \\ &= \frac{(-1)^n \sinh 1}{(i n \pi + 1)} \\ &= \frac{(-1)^n \sinh 1 (-i n \pi + 1)}{(i n \pi + 1)(-i n \pi + 1)} \end{aligned}$$

$$\Rightarrow c_n = \frac{(-1)^n \sinh 1 (-in\pi + 1)}{(n^2\pi^2 + 1)}$$

Therefore, the complex form of Fourier Series for $f(x)$ is

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/l}$$

$$\Rightarrow e^{-x} = \sum_{n=-\infty}^{\infty} \frac{(-1)^n \sinh 1 (-in\pi + 1)}{(n^2\pi^2 + 1)} e^{in\pi x}$$

REFERENCES:

1. Higher Engineering Mathematics-B.S. Grewal
2. Higher Engineering Mathematics-B.V. Ramana
